Richard Brue

Computer Engineer

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Professional Summary

I am a driven software developer with over six years of experience in programming, technical problem-solving, and system design. My background includes C, C++, Python, Java, JavaScript, SQL, and assembly, with hands-on work in embedded systems, application development, and database solutions. I have contributed to projects in renewable energy, autonomous vehicles, and mobile apps. I am especially interested in firmware and system design, with a long-term goal of building technology for EV charging and sustainable energy. I look to be Proficient in designing and implementing firmware for IoT devices and embedded systems, I have successfully enhanced system performance and reliability in various high-impact projects. My expertise in system architecture and project management has led to significant improvements in energy efficiency and data processing capabilities. Recognized for strong collaboration skills and the ability to work seamlessly with cross-functional teams, I am committed to driving technological advancements that support sustainable development.

Work Experience

Bartender / Server | Von's 1000 Spirits

1225 1st Ave Seattle, Wa April 2019

- Fast-paced service, strong teamwork, high-volume shifts
- Excellent customer service skills were key components of my role as a Bartender/Server at Von's 1000 Spirits.
- Honed my ability to manage multiple tasks efficiently.
- Maintained a positive and engaging atmosphere for guests.
- Developed strong communication and interpersonal skills.
- Built rapport with customers to ensure a memorable dining experience.
- Enhanced problem-solving abilities.

Learned the importance of adaptability in a dynamic environment.

Expo / Busser — Dimitriou's Jazz Alley

Seattle, Wa **Feb 2018 – Sept 2018**

- Ensured timely delivery of dishes to guests
- Maintained cleanliness and organization in the dining area
- Collaborated with waitstaff and kitchen team to improve service efficiency
- Enhanced guest satisfaction through effective teamwork
- Demonstrated strong attention to detail

Worked efficiently under pressure during high-volume shifts

Lead Line Cook — Duke's La Jolla

La Jolla, CA **|Apr 2016 – Jan 2018**

- Led kitchen team and maintained quality standards
- Developed and implemented new menu items, focusing on Hawaiian flavours and seasonal ingredients
- Trained and mentored junior staff, promoting a cohesive and efficient kitchen environment
- Managed inventory and supplies, ensuring cost-effective purchasing and minimal waste
- Collaborated with front-of-house staff to ensure seamless service and guest satisfaction

Received positive feedback for consistency and creativity in Hawaiian-inspired dishes

Education

Bachelor of Science in Computer Engineering **Graduated August 2025**

University of Washtington-Tacoma

Focused in System Design - Embedded Systems, Data Structures/Algorhythm, Electrical Systems

Associates of Science in Engineering **Graduated May 2022**

Seattle Central College

Focused in Engineering- Calculus based physics series and Circuit

Technical Skills

- Programming: C, C++, Python, Java, JavaScript, SQL, Assembly
- System Development: Firmware, embedded systems, system design
- Database Tools: SQL, relational database design, optimization
- Mobile: Android development (published on Google Play)
- Hardware: Electrical systems, hardware interfacing
- Other: 3D modelling, troubleshooting, documentation

References

- Josh Goure — JGOURE@YAHOO.COM
- Mitch Brown, MBA, PMP — 440-487-7136
- George Dubuque — 920-559-9166

Projects

Rideshare App

Built a rideshare app focused on safety, user control, and smooth communication.

- Verification for drivers and passengers
- User preferences for matching
- Dedicated interfaces for drivers, passengers, and admins
- Firebase for real-time data and location tracking
- SQL for structured data storage
- In-app communication tools

Progress so far:

- Location module for real-time tracking
- Ride search and map display
- Models for User, Driver, and Ride
- Early safety features and verification groundwork
- Beginning layouts for driver, user, and admin views

Next steps:

- Add advanced verification
- Complete ride-request flow and payment integration
- Build full chat and call features
- Improve speed and stability
- Prepare for beta testing and feedback

linkedin

https://github.com/RichiebM3

Charging Buoy

- Contributed to the development of a renewable-energy buoy for marine transportation, with a focus on embedded programming and hardware integration.
- Designed software algorithms to optimize energy capture and storage, enhancing buoy efficiency under varied sea conditions.
- Collaborated with a multidisciplinary team to create robust hardware solutions, integrating sensors and communication systems for real-time monitoring and control.
- The buoy utilizes induction properties to convert wave motions into electricity, aiming to harness ocean waves for clean energy and reduce the carbon footprint of marine vessels.

Advanced sustainable marine technology and gained valuable insights into renewable energy potential.

Destiny Data Damage

- Developed a DPS simulator utilizing SQL and Java to replicate complex game mechanics.
- Enabled user input for character stats, weapon options, and skill sets to estimate damage outputs in diverse scenarios.
- Designed the tool to enhance understanding of individual elements' impact on overall performance and optimize gameplay strategies.
- Integrated real-time data analysis and an intuitive interface to provide actionable insights for battle efficiency, upgrade planning, and improved team coordination.

Autonomous Hexapod Platform

- A fully automated six-legged robotic system designed for advanced mobility and adaptability in various terrains.
- Self-Regulating Six-Legged Mechanism: An intelligent mechanism featuring self-adjusting capabilities to maintain balance and stability during operation.
- Independent Hexapod Framework: A standalone structural design enabling autonomous functionality without external control dependencies.
- Self-Operating Six-Limbed System: A six-limbed robotic system capable of executing tasks independently, with integrated sensors and control algorithms.
- Self-Sustaining Hexapod Structure: A robust and durable hexapod design engineered to operate continuously with minimal external intervention.